

Chapter 1

Section 8 - Advanced Counting Techniques

1.1 Section 8.1 - Generating Functions

How do we determine an explicit formula for a recursive sequence?

For example,

$$\begin{aligned}f_0 &= 0 \\f_1 &= 1 \\f_{n+1} &= f_n + f_{n-1}, \quad n \geq 1\end{aligned}$$

Has an explicit function of

$$f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

Let's consider a nicer example

$$\begin{aligned}a_0 &= 6 \\a_1 &= 13 \\a_n &= 5a_{n-1} - 6a_{n-2}, \quad n \geq 2\end{aligned}$$

Let's write out the first few terms

$$\begin{aligned}a_2 &= 5a_1 - 6a_0 \\&= 5 \cdot 13 - 6 \cdot 5 \\&= 29\end{aligned}$$

$$\begin{aligned}a_3 &= 5 \cdot 29 - 6 \cdot 13 \\&= 67\end{aligned}$$

$$\begin{aligned}a_4 &= 5 \cdot 67 - 6 \cdot 29 \\&= 161\end{aligned}$$

We form a **generating function** for the sequence

$$a_0, a_1, a_2, \dots$$

Which is a power series whose coefficients are the sequence

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

From our example above, the generating function is

$$f(x) = 6 + 13x + 29x^2 + 67x^3 + 161x^4 + \dots$$

To reduce this, let's go back to our recursive function

$$a_n = 5a_{n-1} - 6a_{n-2}$$

Rearranging, we get

$$a_n - 5a_{n-1} + 6a_{n-2} = 0$$

Now let's bring in our generating function

$$\begin{aligned}f(x) &= 6 + 13x + 29x^2 + 67x^3 + 161x^4 + \dots \\-5xf(x) &= -30x - 65x^2 - 145x^3 - 335x^4 - \dots \\6x^2f(x) &= 36x^2 + 78x^3 + 174x^4 + \dots\end{aligned}$$